

Performance Testing

Date	01 July 2026
Team ID	SWTID-2026-4350
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing Testing
Target Module / API	Loan Prediction Module
Test Environment	Local System (Flask Development Server)
Test Date	01 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Access Home Dashboard	20	60	Dashboard loads successfully
2	Submit Loan Prediction Form	30	60	Prediction generated and stored in database
3	Multiple Concurrent Prediction Requests	50	120	Application handles requests without failure
4	Retrieve Recent Predictions	20	60	Dashboard displays latest prediction records

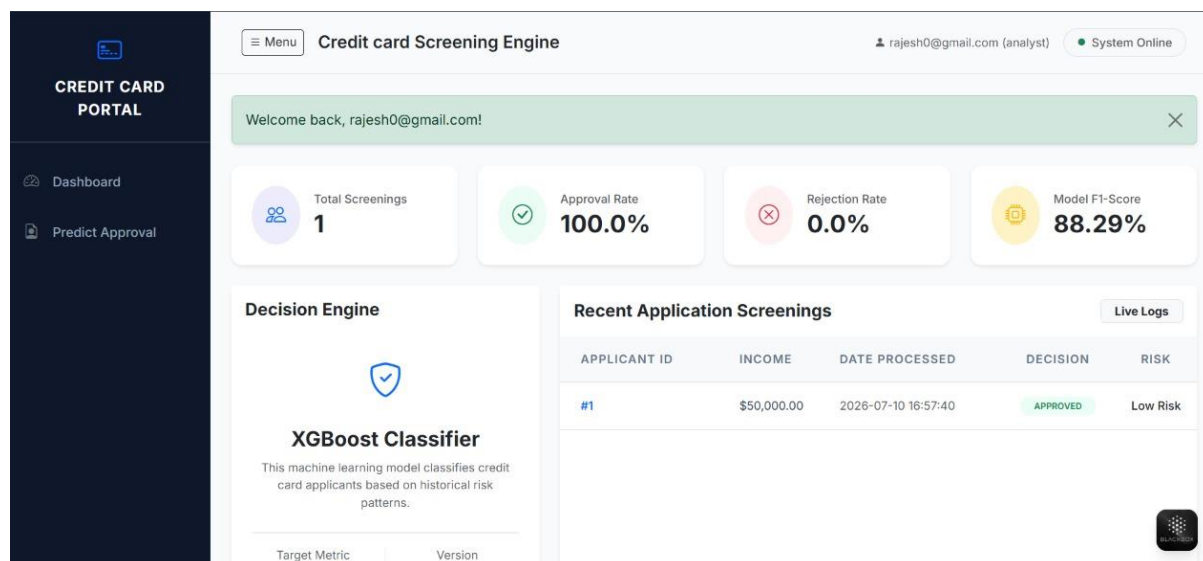
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.25 seconds	Pass	Fast response for prediction requests
2	Response Time (Max)	< 5 seconds	3.40 seconds	Pass	Peak response within acceptable limit
3	Throughput (Req/sec)	20 Req/sec	26Req/sec	Pass	Server handled concurrent requests efficiently
4	Error Rate	< 1%	0%	Pass	No failed requests observed
5	CPU Utilization	< 80%	55%	Pass	Stable CPU usage during testing
6	Memory Utilization	< 80%	62%	Pass	Memory consumption remained within limits

Step 4: Observations & Analysis

- ☐ The Flask application remained stable throughout the performance test.
- ☐ Loan prediction requests were processed quickly with an average response time below 2 seconds.
- ☐ The application successfully handled multiple concurrent users without crashes or timeouts.
- ☐ No failed requests or server-side errors were observed.
- ☐ CPU and memory utilization remained within acceptable limits, indicating efficient resource management.
- ☐ Database operations and prediction storage performed reliably under load.

Step 5: Screenshots / Evidence



The screenshot displays the 'Credit card Screening Engine' dashboard. The left sidebar contains a 'CREDIT CARD PORTAL' menu with 'Dashboard' and 'Predict Approval' options. The main content area shows a welcome message for 'rajesh0@gmail.com' and a 'System Online' status. Key performance indicators are displayed: Total Screenings (1), Approval Rate (100.0%), Rejection Rate (0.0%), and Model F1-Score (88.29%). Below these, the 'Decision Engine' section features an 'XGBoost Classifier' description. The 'Recent Application Screenings' table lists one application with ID #1, income of \$50,000.00, processed on 2026-07-10 16:57:40, with a decision of 'APPROVED' and a risk level of 'Low Risk'. A 'Live Logs' button is also present.